Practical 1

Aim: Study of Python Basic Libraries such as Statistics, Math, Numpy and Scipy

Source-Code and Output:

1. **Statistics**

```
import statistics
data=input("Enter data separated by comma:")
data=[int(x) for x in data.split(',')]
mean=statistics.mean(data)
median=statistics.median(data)
mode=statistics.mode(data)
stdev=statistics.stdev(data)
variance=statistics.variance(data)
print(f"mean={mean}")
print(f"median={median}")
print(f"mode={mode}")
print(f"Standard Deviation={stdev}")
print(f"variance={variance}")
```

```
Enter data separated by comma:11,22,33,44,55
mean=33
median=33
mode=11
Standard Deviation=17.392527130926087
variance=302.5
```

2. **Math**

```
import math
x=int(input("Enter a number:"))
result1=math.ceil(x)
result2=math.factorial(x)
result3=math.exp(x)
result4=math.sqrt(x)
result5=math.floor(x)
n1=int(input("Enter first number:"))
n2=int(input("Enter second number:"))
result6=math.gcd(n1,n2)
result7=math.pow(n1,n2)
angle=int(input("Enter an angle:"))
a=math.radians(angle)
result8=math.sin(a)
result9=math.cos(a)
result10=math.tan(a)
print(f"Using math.ceil({x})={result1}")
print(f"Using math.factorial({x})={result2}")
print(f"Using math.exp({x})={result3}")
print(f"Using math.sqrt({x})={result4}")
print(f"Using math.floor({x})={result5}")
print(f"Using math.gcd({n1}, {n2})={result6}")
print(f"Using math.pow({n1}, {n2})={result7}")
print(f"Using math.sin({angle} degrees)={result8}")
print(f"Using math.cos({angle} degrees)={result9}")
print(f"Using math.tan({angle} degrees)={result10}")
```

```
Enter a number:22
Enter first number:11
Enter second number:33
Enter an angle:45
Using math.ceil(22)=22
Using math.factorial(22)=1124000727777607680000
Using math.exp(22)=3584912846.131592
Using math.sqrt(22)=4.69041575982343
Using math.floor(22)=22
Using math.gcd(11, 33)=11
Using math.pow(11, 33)=2.3225154419887806e+34
Using math.sin(45 degrees)=0.7071067811865475
Using math.cos(45 degrees)=0.7071067811865476
Using math.tan(45 degrees)=0.9999999999999999
```

3. **Scipy**

```
from scipy import linalg
import numpy as np
from scipy import special
from scipy import constants
print("Seconds in a minute:", constants.minute)
print("Seconds in an hour:", constants.hour)
print("Length of an inch in meters:", constants.inch)
print("Volume of a liter in cubic meters:", constants.liter)
exp_input = float(input("Enter the exponent for 10^x: "))
print(f"10^{exp_input} = {special.exp10(exp_input)}")
angle=float(input("Enter an angle for sine and cosine(in degrees):"))
print(f"sin({angle}degrees) = {special.sindg(angle)}")
print(f"cos({angle}degrees) = {special.cosdg(angle)}")
rows=int(input("Enter number of rows for matrix:"))
cols=int(input("Enter number of columns for matrix:"))
mat=list(map(float,input(f"Enter {rows*cols} elements for the matrix separated by comma:").split(',')))
mat=np.array(mat).reshape(rows,cols)
if rows==cols:
    x=linalg.det(mat)
    print(f"Determinant of the matrix\n{mat}\n is: {x}")
else:
    print("Determinant can only be calculated for square matrices.")
```

```
Seconds in a minute: 60.0
Seconds in an hour: 3600.0
Length of an inch in meters: 0.0254
Volume of a liter in cubic meters: 0.001
Enter the exponent for 10^x: 3
10^3.0 = 1000.0
Enter an angle for sine and cosine(in degrees):45
sin(45.0degrees) = 0.7071067811865476
cos(45.0degrees) = 0.7071067811865475
Enter number of rows for matrix:2
Enter number of columns for matrix:2
Enter 4 elements for the matrix separated by comma:9,8,7,6
Determinant of the matrix
[[9. 8.]
 [7. 6.]]
 is: -1.999999999999993
```

4. **Numpy**

```
import numpy as np
rows=int(input("Enter number of rows:"))
cols=int(input("Enter number of columns:"))
elements=list(map(float,input(f"Enter {rows*cols} elements separated by comma:")
.split(',')))
user_array=np.array(elements).reshape(rows,cols)
ones_array=np.ones((rows,cols))
zeros_array=np.zeros((rows,cols))
transposed_array=np.transpose(user_array)
reshaped_array=user_array.reshape(rows * cols)
min_value=np.min(elements)
max_value=np.max(elements)
print("User Array:\n",user_array)
print("Ones Array:\n",ones_array)
print("Zeros Array:\n",zeros_array)
print("Transposed Array:\n",transposed_array)
print("Reshaped Array(Flattened):\n",reshaped_array)
print(f"Minimum value in the array: {min_value}")
print(f"Maximum value in the array: {max_value}")
```

```
Enter number of rows:2
Enter number of columns:2
Enter 4 elements separated by comma:2,3,4,5
User Array:
 [[2. 3.]
 [4. 5.]]
Ones Array:
 [[1. 1.]
 [1. 1.]]
Zeros Array:
 [[0. 0.]
 [0. 0.]]
Transposed Array:
 [[2. 4.]
 [3. 5.]]
Reshaped Array(Flattened):
 [2. 3. 4. 5.]
Minimum value in the array: 2.0
Maximum value in the array: 5.0
```